

Bayesian Ranking and Graph-Based Review Design under Sparse Evaluation Data

A Case Study of the CRS Seed Grants 2026

Victor Sieber

BSc Mathematics student

Department of Mathematics, University of Zurich

Supervised by Dr. Rachel Heyard

2026-08-20

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Abstract

1 Introduction

1.1 Grant evaluation under limited information

Grant peer review converts a limited number of expert judgments into decisions that can have substantial consequences for applicants. The challenge is not only disagreement between reviewers, but also the risk of overinterpreting small score differences when each proposal is assessed by only a few reviewers. Kaplan et al. show directly that finer scoring precision requires increasingly many independent reviewers, and illustrate how rankings can vary strongly when only a small number of evaluations are available ([Kaplan et al. 2008](#)).

This uncertainty becomes particularly important when proposals are close in score or close to a decision boundary. In a retrospective analysis of Australian grant panels, Graves et al. found substantial instability in funding status after accounting for variability among panel members' scores ([Graves et al. 2011](#)). Jayasinghe, Marsh and Bond reported low single-rater reliability in Australian Research Council grant assessments and used a multilevel cross-classified model to separate proposal- and assessor-related sources of variation ([Jayasinghe et al. 2003](#)). Heyard et al. similarly motivate Bayesian ranking by low inter-rater reliability and by the difficulty of distinguishing proposals of similar quality near a funding line ([Heyard et al. 2022](#)). These findings do not imply that peer review is uninformative; rather, they motivate distinguishing an observed numerical ordering from the strength of evidence supporting that ordering.

The present project studies this distinction in a small evaluation setting. With only two reviews per proposal, a method that reports a single rank without its uncertainty can appear more decisive than the available information warrants. The aim is therefore to model proposal and reviewer variation explicitly, quantify posterior uncertainty in the resulting ranks, and later ask whether the structure of the review design itself can indicate where additional review effort would be most informative.

1.2 CRS Seed Grant evaluation 2026

The Center for Reproducible Science and Research Synthesis (CRS) at the University of Zurich (UZH) works to advance scientific methods and practices that support trustworthy and reproducible research. Its activities include research on reproducibility, research synthesis, and meta-research ([Center for Reproducible Science and Research Synthesis 2026b](#)).

In 2026, the CRS launched its first Seed Grant call to support early-stage projects and initiatives in research synthesis, meta-research, and related fields ([Center for Reproducible Science and Research Synthesis 2026b](#)). The evaluation round included 42 proposals assessed by a panel of nine reviewers. Each proposal

received exactly two independent reviews, resulting in 84 completed review records.

Reviewers evaluated each proposal on four supporting criteria: scientific quality; potential for scaling and obtaining substantial third-party funding; relevance to fostering a sustainable, innovative, and diverse Swiss “research on research” community; and commitment to good research practices, open science, and reproducibility. The original call described selection using the average score across these criteria and reviewers ([Center for Reproducible Science and Research Synthesis 2026b](#)). During the development of the evaluation procedure, this was subsequently replaced by a separate holistic overall grade on a five-point scale, which became the score used for qualification. The reason for this change was not documented in the material available for this project.

The call indicated that approximately six projects could be funded ([Center for Reproducible Science and Research Synthesis 2026b](#)). In the final evaluation procedure, four proposals met the qualification rule and were selected for funding, as documented on the published results page ([Center for Reproducible Science and Research Synthesis 2026a](#)). The rule is reconstructed explicitly in Section 2.3.

1.3 Motivation for Bayesian ranking

The final CRS qualification procedure was transparent and easy to apply, but it did not quantify uncertainty around the observed overall grades. In particular, two proposals with similar score pairs can appear distinctly ordered by their observed mean grades even when the evidence that one is better than the other is weak. With only two reviews per proposal, the observed scores also reflect both proposal-related variation and differences in reviewer scoring behaviour.

Bayesian hierarchical modelling provides a way to represent these sources of variation separately within a probabilistic model. Heyard et al. model latent proposal effects together with assessor-related effects and rank proposals using the posterior distribution of the proposal effects rather than the raw score averages alone ([Heyard et al. 2022](#)). Their expected-rank approach is designed to retain information about uncertainty instead of treating an estimated ordering as known with certainty.

For the CRS data, the Bayesian analysis is therefore used to support decisions and quantify uncertainty rather than as a replacement rule that mechanically determines funding.

1.4 Research questions

1.4.1 Research Question 1

What ranking does the Bayesian model produce after accounting for reviewer differences, and how uncertain are the resulting ranks?

1.4.2 Research Question 2

How does the main Bayesian ranking model compare with the final CRS qualification rule and the ranking based on the observed overall grades?

1.4.3 Research Question 3

Is the main Bayesian model computationally stable, adequate under posterior predictive checks, and robust to the tested prior specification?

1.4.4 Research Question 4

How closely do the four supporting criterion scores correspond to the holistic overall grade, and would an equal-weight average of the criteria have produced a materially different proposal ordering?

1.4.5 Research Question 5

Can the sparse reviewer–proposal design be used to identify where additional review effort may be most informative?

1.5 Scope and structure of the report

2 Evaluation design and descriptive analysis

2.1 Evaluation process and analytical data

The pseudonymized analytical dataset contains 84 completed reviews of 42 proposals by nine reviewers. Each proposal received exactly two independent reviews. Reviewer workloads ranged from 6 to 12 reviews because assignments were based on expertise and availability.

Each review contains four supporting criterion scores and a separate holistic overall grade. All scores use an integer scale from 1 to 5, where higher values indicate a more favourable evaluation. The Bayesian models use the holistic overall grade as their outcome. The four criterion scores are retained for a separate descriptive analysis of their relationship with the overall grade.

2.2 Reviewer–proposal assignment

The reviewer–proposal design is sparse and unbalanced. Each proposal was reviewed twice, whereas reviewers assessed different numbers of proposals. Assignments were not random but depended on expertise and availability.

Consequently, estimated reviewer effects should be interpreted as scoring tendencies rather than direct evidence of reviewer bias. Differences attributed to reviewers may partly reflect differences in the types of proposals assigned to them.

2.3 Reconstruction of the final qualification procedure

According to the published CRS Seed Grant results ([Center for Reproducible Science and Research Synthesis 2026a](#)), funding was available for proposals that received the highest overall grade from at least one reviewer and no grade below the second-highest category. Since the overall grade was scored on a scale from 1 to 5, this means that a proposal qualified if it received at least one grade of 5 and its second grade was at least 4. With exactly two reviews per proposal, the qualifying score combinations were therefore (5,5), (5,4), and (4,5). Equivalently, a proposal qualified if its mean overall grade was at least 4.5.

This procedure should be interpreted as a score-based qualification threshold rather than as a fixed top-four selection rule. Four proposals satisfied the threshold in the 2026 evaluation round. Since the call indicated that approximately six projects could be funded, the available funding capacity was not binding and no additional prioritization among qualified proposals was required.

2.4 Criteria and holistic overall grade

Previous NIH analyses have examined how criterion scores relate to holistic overall evaluations ([Eblen et al. 2016](#); [Erosheva et al. 2020](#)). Eblen et al. found substantial differences between criteria in their association with the Overall Impact score ([Eblen et al. 2016](#)). This motivates examining whether the four supporting CRS criteria correspond equally closely to the separately assigned holistic overall grade.

Let C_{rc} denote the score assigned in review r to criterion c , and let Y_r denote the holistic overall grade. Their relationship was summarized descriptively using Pearson correlation for linear association and Spearman correlation for monotonic association based on ranks. The corresponding sample definitions are given in [Section 9.1](#). These correlations are treated descriptively because the 84 reviews are not independent with respect to proposals and reviewers.

Criterion	Mean criterion score	Pearson correlation	Spearman correlation
Scientific quality	3.643	0.892	0.890
Community building	3.179	0.545	0.551
Scaling potential	3.512	0.541	0.553
Scientific rigor	3.881	0.588	0.599

Scientific quality showed the strongest relationship with the holistic overall grade, with Pearson and Spearman correlations of 0.892 and 0.890. The remaining three criteria were positively related to the overall grade but less strongly, with

correlations around 0.54–0.60. These correlations do not, however, identify the weights reviewers placed on the individual criteria.

For a simple equal-weight summary, define the criterion average for review r as

$$A_r = \frac{1}{4} \sum_{c=1}^4 C_{rc}.$$

Across the 84 reviews, the mean holistic overall grade was 3.476 and the mean equal-weight criterion average was 3.554. The Pearson correlation between A_r and Y_r was 0.859 and the Spearman correlation was 0.849. The mean difference $A_r - Y_r$ was 0.077 grade points, indicating little systematic difference in level between the two summaries, and 78.6% of reviews had a criterion average within 0.5 grade points of the holistic overall grade.

2.5 Equal-weight criterion summary at proposal level

The equal-weight summary is used as a descriptive counterfactual rather than as an estimate of how reviewers actually combined the four criteria. For proposal i , let

$$O_i = \frac{1}{2} \sum_{r:p(r)=i} Y_r$$

denote its mean holistic overall grade, and let

$$A_i = \frac{1}{2} \sum_{r:p(r)=i} A_r = \frac{1}{8} \sum_{r:p(r)=i} \sum_{c=1}^4 C_{rc}$$

denote the equal-weight average of all eight criterion scores received across its two reviews.

The two proposal-level summaries remained strongly related, with Pearson correlation 0.848 and Spearman correlation 0.881. Their relationship is shown in Figure 1.

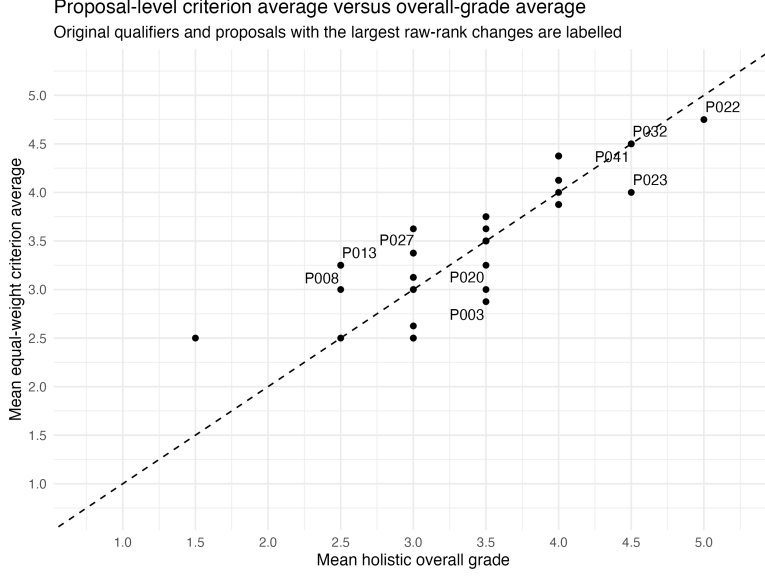


Figure 1: Proposal-level equal-weight criterion average versus mean holistic overall grade. The dashed identity line represents identical proposal-level summaries. CRS qualifiers and proposals with the largest **raw-rank changes** are labelled.

The holistic proposal average is relatively coarse because it is calculated from only two integer-valued grades, whereas A_i averages eight criterion scores and therefore has finer numerical resolution. Raw-rank changes are consequently sensitive to the breaking of ties and are not emphasized here.

At the top of the distribution, P022 had criterion average 4.75 and P032 and P041 had 4.50. These are three of the four proposals that qualified under the final CRS rule. The fourth qualifier, P023, had criterion average 4.00, while several non-qualified proposals had criterion averages of 4.375. Thus, equal weighting preserves substantial agreement with the holistic evaluation but does not reproduce its upper ordering exactly. Since equal weighting was not the CRS decision rule, this comparison is interpreted as a counterfactual rather than as an alternative reconstruction of the final qualification procedure.

3 Bayesian ranking framework

3.1 Related work

The Bayesian ranking framework used in this project follows directly from Heyard et al. (2022), who developed a Bayesian hierarchical model for the evaluation of research proposals at the Swiss National Science Foundation. Their model represents observed evaluation scores through latent proposal and assessor-related

effects. Rather than ranking proposals only by observed scores or posterior mean proposal effects, they use the posterior distribution of the proposal effects to calculate expected ranks. This expected-rank formulation, which accounts for uncertainty in the relative ordering of proposals, is also the principal Bayesian ranking quantity used in this report.

Heyard et al. place this approach within an earlier literature on Bayesian ranking. In particular, they draw on Laird and Louis (1989), who developed Bayes and empirical-Bayes methods for estimating the ranks of random effects in variance-component models. Heyard et al. use their expected-rank formulation,

$$E[\text{rank}(\theta_i)],$$

as the basis for ranking the latent proposal effects.

Bayesian hierarchical modelling has also previously been applied directly to grant-review ratings. Johnson (2008) analysed NIH peer-review ratings using a Bayesian hierarchical model that allows for differences in reviewer scoring patterns and quantifies posterior uncertainty in proposal ratings. For ordinal grant scores, Heyard et al. additionally draw on Johnson (2008) and Cao, Stokes and Zhang (2010) to formulate an alternative model based on an underlying continuous proposal-quality trait that is translated into observed ordinal categories. Cao et al. develop such a Bayesian hierarchical multirater model explicitly for ranking, allowing both a latent item trait and rater-specific characteristics, and demonstrate the approach using grant-review data.

The proposal-reviewer structure is also related to the cross-classified grant-review model of Jayasinghe, Marsh and Bond (2003). When reviewers assess multiple proposals and proposals receive evaluations from multiple reviewers, proposal and reviewer effects are crossed rather than nested. This provides the structural motivation for retaining persistent proposal and reviewer effects in the simpler V1 specification used here.

More recently, Heyard et al. (2025) applied Bayesian ranking to individual evaluations of 1,006 Marie Skłodowska-Curie Actions proposals and compared the resulting model-based rankings with scores, rankings and funding decisions established after consensus meetings. Their analysis again used expected ranks of latent proposal effects while adapting the hierarchical model to the information available on individual experts.

The present analysis therefore does not introduce a new Bayesian ranking method. It takes the framework of Heyard et al. as its methodological starting point, reproduces their continuous reference specification as V0, and examines whether a simpler crossed proposal-reviewer specification preserves the substantive ranking conclusions in the CRS setting.

3.2 V0: Heyard reference model

The reference model V0 reproduces the continuous homogeneous-variance specification of Heyard et al. (2022). Let $r = 1, \dots, 84$ index completed reviews, let $p(r)$ denote the proposal reviewed in record r , and let $a(r)$ denote the reviewer. The observed holistic overall grade is Y_r , and $\bar{Y}$ denotes its observed mean.

The model is

$$Y_r \mid \theta, \lambda \sim \mathcal{N}(\bar{Y} + \theta_{p(r)} + \lambda_{p(r), a(r)}, \sigma^2),$$

with proposal effects

$$\theta_i \sim \mathcal{N}(0, \tau_\theta^2),$$

and proposal–reviewer assessment effects

$$\lambda_{ij} \sim \mathcal{N}(\nu_j, \tau_\lambda^2), \quad \nu_j \sim \mathcal{N}(0, 0.5^2).$$

The latent proposal effect θ_i is the quantity used for ranking. The reviewer-specific mean ν_j represents a persistent scoring tendency, while $\lambda_{ij} - \nu_j$ permits an additional deviation for a particular proposal–reviewer assessment.

V0 is retained as the methodological reference model rather than as the preferred CRS specification. Each observed proposal–reviewer combination occurs only once in the CRS data, so the additional assessment-specific effect is weakly informed at the individual level. This motivates the simpler V1 formulation.

3.3 V1: CRS cross-classified model

The CRS-specific model V1 removes the proposal–reviewer-specific assessment term from V0 and retains one persistent proposal effect and one persistent reviewer effect. For review r ,

$$Y_r \sim \mathcal{N}(\bar{Y} + \theta_{p(r)} + b_{a(r)}, \sigma^2),$$

where

$$\theta_i \sim \mathcal{N}(0, \tau_\theta^2), \quad b_j \sim \mathcal{N}(0, \tau_b^2).$$

Equivalently,

$$Y_r = \bar{Y} + \theta_{p(r)} + b_{a(r)} + \varepsilon_r, \quad \varepsilon_r \sim \mathcal{N}(0, \sigma^2).$$

The proposal effect θ_i is again the latent quantity used for ranking. The reviewer effect b_j is shared across all proposals assessed by reviewer j and represents a reviewer-associated scoring tendency. Because reviewer assignments were based on expertise and availability rather than random allocation, these effects are interpreted descriptively rather than as causal measures of reviewer bias.

The model is cross-classified because each observation belongs simultaneously to a proposal and a reviewer, with neither grouping nested within the other. The hierarchical specification also induces partial pooling: proposal effects are shrunk toward the overall proposal mean according to the estimated between-proposal variation τ_θ^2 , and reviewer effects are pooled analogously through τ_b^2 . This regularization is particularly relevant in the CRS setting because each proposal contributes only two observed grades.

V1 is used as the main CRS model because it represents the recurring proposal and reviewer structure directly while avoiding a separate latent effect for every one-off proposal-reviewer assessment.

3.4 Assumptions and prior specification

Both V0 and V1 treat the observed grades as conditionally Gaussian with a common residual variance and assume exchangeable proposal effects around zero. V1 additionally treats the reviewer effects as exchangeable around zero. The overall mean $\bar{Y}$ is fixed at the observed mean grade rather than estimated as a separate intercept.

For V0, the scale parameters use

$$\sigma, \tau_\theta, \tau_\lambda \sim \text{Uniform}(0, 2),$$

while V1 uses

$$\sigma, \tau_\theta, \tau_b \sim \text{Uniform}(0, 2).$$

The lower endpoint is implemented numerically as 10^{-6} . The sensitivity of the main V1 conclusions to this scale-prior specification is examined in Chapter 5.

Although the holistic overall grade is recorded on an integer scale from 1 to 5, both models use the continuous Gaussian formulation of Heyard et al. (2022). This treats adjacent categories as equally spaced. Since the main target is the relative ranking of latent proposal effects rather than prediction of valid score categories, this approximation is retained here; its limitations are assessed through posterior predictive checks in Chapter 5.

3.5 Posterior ranking quantities

The ranking is based on the posterior distribution of the proposal effects $\theta_1, \dots, \theta_{42}$ rather than directly on the observed score averages. For posterior draw s , the rank of proposal i is

$$R_i^{(s)} = 1 + \sum_{k \neq i} \mathbb{1}_{\{\theta_k^{(s)} > \theta_i^{(s)}\}},$$

where rank 1 corresponds to the proposal with the largest latent proposal effect and $\mathbb{1}_{\{\cdot\}}$ denotes the indicator function.

The posterior mean of this rank distribution is the expected rank,

$$ER_i = E[R_i \mid Y].$$

Using the definition of R_i ,

$$ER_i = 1 + \sum_{k \neq i} \mathbb{P}(\theta_k > \theta_i \mid Y).$$

Expected rank can therefore be interpreted as 1 plus the sum of the posterior probabilities that each competing proposal has a larger latent effect than proposal i . In practice, it is estimated from the posterior draws by

$$ER_i \approx \frac{1}{S} \sum_{s=1}^S R_i^{(s)}.$$

Expected rank is the principal Bayesian ranking quantity used in this report and follows Heyard et al. (2022). With 42 proposals, the midpoint of the rank scale is

$$\frac{42 + 1}{2} = 21.5.$$

When many of the pairwise ordering probabilities for proposal i are close to 0.5 ($\mathbb{P}(\theta_k > \theta_i \mid Y) \approx 0.5$), its expected rank is consequently pulled toward the centre of the rank scale.

Three ranking summaries are distinguished throughout the analysis. The **raw-score rank** orders proposals by their mean observed overall grade, the **posterior-mean rank** orders the posterior means of θ_i , and the **expected rank** averages the proposal's rank across the full posterior distribution. Rank uncertainty is summarized by the 90% posterior rank interval, defined by the 5th and 95th percentiles of R_i .

For comparisons between particular proposals, pairwise posterior probabilities such as

$$\mathbb{P}(\theta_i > \theta_k \mid Y)$$

give a more direct measure of the evidence for one proposal being ranked above another. These are particularly useful when small changes in a model produce different integer ranks for proposals that are nearly tied.

3.6 Posterior computation and diagnostics

All Bayesian models were fitted in JAGS from R. V0 uses the explicit Heyard reference specification implemented through `ERforResearch`, while V1 uses a custom JAGS model with one persistent reviewer effect per reviewer.

Unless stated otherwise, each final fit used four Markov chains, 10,000 adaptation iterations, 10,000 burn-in iterations and 50,000 retained sampling iterations per chain, giving 200,000 posterior draws in total. All MCMC fits used a fixed random seed of 20260727 to ensure reproducibility.

Posterior-mean ranks, expected ranks, rank intervals and pairwise posterior probabilities were calculated from the retained posterior draws. Numerical convergence was assessed using the potential scale reduction factor (PSRF) and effective sample size (ESS). These diagnostics assess whether the posterior computation is stable; statistical model adequacy is evaluated separately through posterior predictive checks and sensitivity analysis in Chapter 5.

4 Bayesian ranking results

4.1 Main V1 ranking and comparison with the final CRS procedure

The CRS-specific V1 model preserves the broad upper ordering based on the observed overall grades while showing that the exact latent ranks are highly uncertain. P022 had posterior-mean rank 1 and expected rank 13.08. P032, P023 and P041 occupied posterior-mean ranks 2, 3 and 4, with expected ranks 15.87, 15.87 and 16.22, respectively. These are exactly the four proposals that satisfied the final CRS qualification threshold.

The posterior-mean ordering is therefore much sharper than the posterior evidence supporting the exact relative positions. The 90% posterior rank interval for P022 was 1–36; for P023 and P032 it was 1–37; and for P041 it was 1–38. The relationship between the observed-score rank, posterior-mean rank and posterior expected rank is shown in Figure 2.

This distinction is the main additional information provided by the Bayesian ranking relative to the final qualification rule. The CRS procedure identifies

proposals that satisfy a transparent score threshold; it does not quantify how strongly the available reviews separate their latent proposal effects. In the present data, the same four proposals appear strongest under both approaches, but their exact positions are weakly identified. The Bayesian analysis is therefore interpreted as uncertainty-aware decision support rather than as a retrospective replacement for the CRS rule.

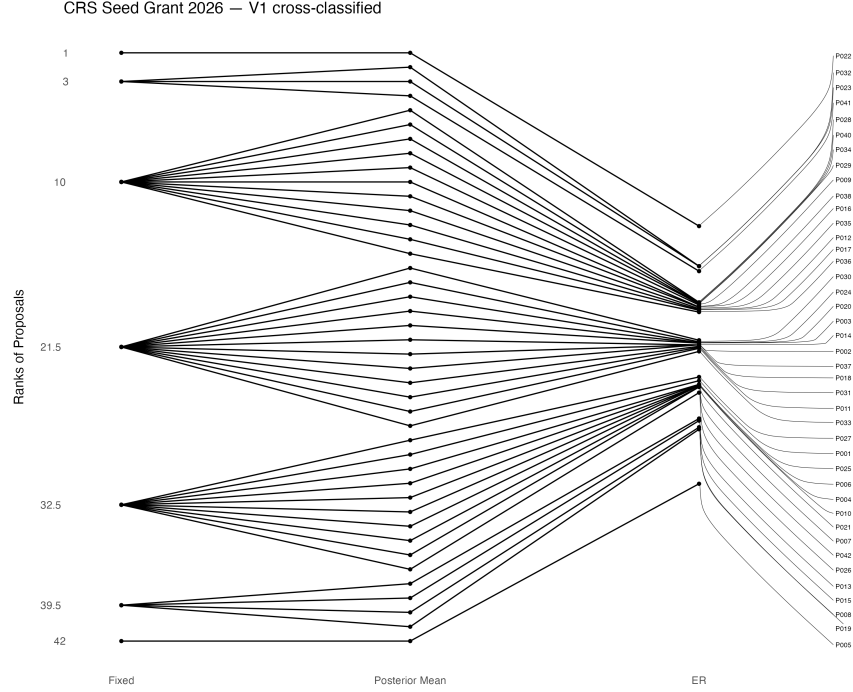


Figure 2: Proposal rankings under the observed-score ranking, posterior-mean ranking and posterior expected rank for the V1 cross-classified model. The movement of expected ranks toward the centre of the rank scale reflects substantial posterior uncertainty in the exact proposal ordering.

4.2 Robustness to the Heyard reference specification

The V0 Heyard reference model gives almost the same substantive result. The four CRS qualifiers again occupy the first four posterior-mean positions. Across all 42 proposals, the Spearman correlation between V0 and V1 posterior-mean ranks was 0.9951 and the Pearson correlation between expected ranks was 0.9916. The close agreement is visible in Figure 3.

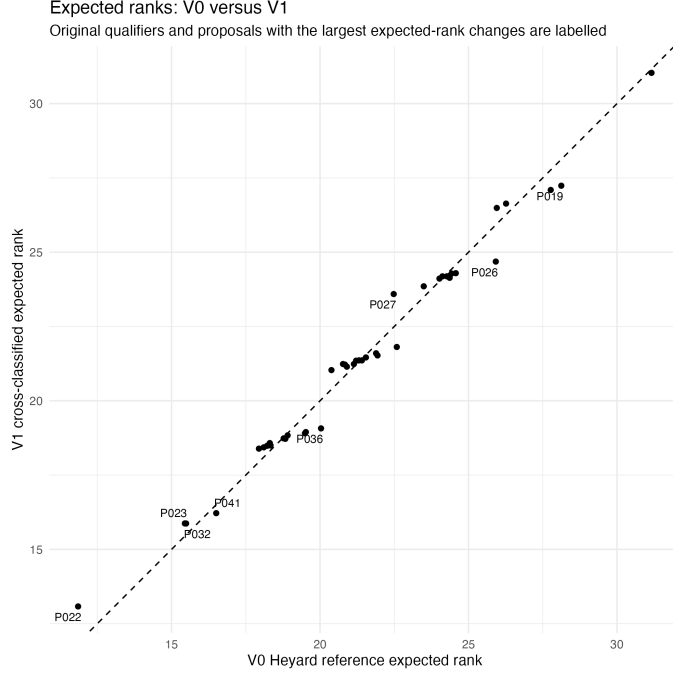


Figure 3: Posterior expected ranks under the V0 Heyard reference model and the V1 cross-classified model. The dashed identity line represents identical expected ranks under both models.

Small changes in integer ranks can nevertheless occur when proposals are almost indistinguishable. Twenty-six of the 42 proposals changed posterior-mean integer rank, but these changes were small: the mean absolute change was 0.86 positions and the maximum was three positions. P023 and P032 provide a clear example: V0 places P023 above P032, while V1 reverses them. Yet

$$\mathbb{P}(\theta_{032} > \theta_{023} \mid Y) = 0.498955$$

under V0 and

$$\mathbb{P}(\theta_{032} > \theta_{023} \mid Y) = 0.500470$$

under V1. The apparent reversal is therefore an almost exact posterior tie rather than evidence that the two models disagree meaningfully about these proposals.

Rank uncertainty is equally stable across the two specifications. The mean width of the 90% posterior rank intervals was 36.50 positions under V0 and 36.74 under V1, with median width 37 under both models. Thus, simplifying V0 to

the cross-classified V1 model changes little about either the broad ordering or the central finding that exact ranks are poorly identified. Detailed V0 ranking output and computational diagnostics are retained in Section 9.2.

4.3 Bayesian ranking using the equal-weight criterion average

The descriptive criterion analysis in Sections 2.4 and 2.5 suggests that the holistic grade and an equal-weight criterion average contain overlapping but not identical information. To examine whether this difference persists after accounting for reviewer-associated scoring tendencies and ranking uncertainty, V1 was fitted a second time using the review-level criterion average A_r as the response:

$$A_r \sim \mathcal{N}\left(\bar{A} + \theta_{p(r)}^{(A)} + b_{a(r)}^{(A)}, \sigma_A^2\right),$$

with

$$\theta_i^{(A)} \sim \mathcal{N}(0, \tau_{\theta, A}^2), \quad b_j^{(A)} \sim \mathcal{N}(0, \tau_{b, A}^2).$$

The reviewer–proposal assignments, priors and MCMC settings were kept identical to the overall-grade V1 fit; only the response and its observed mean changed. The resulting fit had satisfactory computational diagnostics (maximum PSRF 1.0032; minimum ESS 1,863).

Changing the response had a noticeably larger effect than changing from V0 to V1. The Spearman correlation between posterior-mean ranks from the two V1 fits was 0.873, and the Pearson correlation between expected ranks was 0.863. The mean absolute expected-rank change was 2.97 positions and the maximum was 7.89. By comparison, the Pearson correlation between V0 and V1 expected ranks using the same holistic outcome was 0.9916.

The expected-rank comparison is shown in Figure 4.

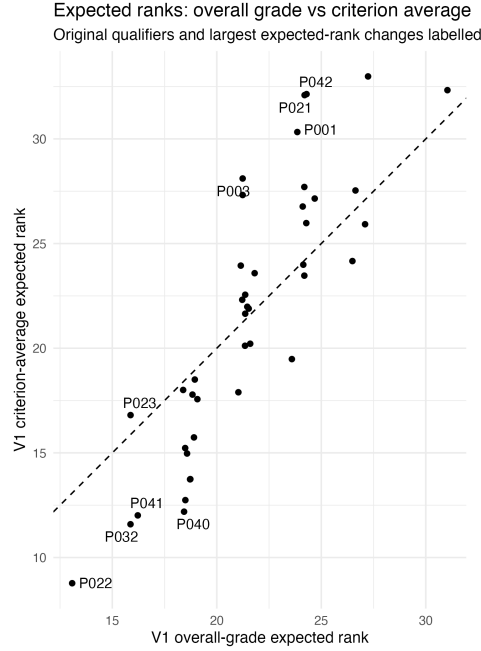


Figure 4: Posterior expected ranks from V1 fitted to the holistic overall grade and to the equal-weight criterion average. The dashed identity line represents identical expected ranks under both response definitions. CRS qualifiers and proposals with the largest expected-rank changes are labelled.

The upper ordering illustrates the difference. P022 and P032 remained at posterior-mean ranks 1 and 2 under the criterion-average model, P041 moved from rank 4 to rank 3, and P040 moved from rank 6 to rank 4. P023 moved from posterior-mean rank 3 to rank 11. That last change is less dramatic when uncertainty is considered: its expected rank changed only from 15.87 to 16.80, whereas P040 improved from 18.43 to 12.19. Expected-rank changes therefore provide a more informative comparison than exact integer ranks alone.

The criterion-average fit also produced somewhat narrower posterior rank distributions. The mean width of the 90% posterior rank intervals decreased from 36.74 to 33.19 positions, while the median width decreased from 37 to 34. The observed criterion average also had lower dispersion than the holistic grade, with standard deviations of 0.772 and 0.950, respectively. Since the criterion average combines four scores, part of this reduction in variation can arise mechanically from averaging. The corresponding change in the fitted variance decomposition is examined in Section 5.2.

The comparison does not establish that one score summary is preferable to the other. It shows that the definition of the evaluation outcome matters materially for the posterior ordering: changing from the holistic overall grade to the equal-

weight criterion average has a larger effect than changing from V0 to V1 while holding the holistic outcome fixed.

5 Validation and sensitivity analysis of V1

5.1 Validation strategy

The close agreement between V0 and V1 shows that the main ranking conclusion is not sensitive to the additional assessment-specific layer in V0, but it does not establish that V1 is an adequate working model. Validation therefore focuses on three questions: whether the posterior computation is stable, whether V1 reproduces important features of the observed grades, and whether the main conclusions depend materially on the baseline scale priors.

5.2 Computational and variance diagnostics

The V1 posterior was sampled reliably. Across all monitored parameters, the maximum PSRF was 1.0017 and the minimum effective sample size was 1,627. The corresponding trace plots are provided in Section 9.3.

To examine how V1 distributes variation across proposals, reviewers and individual reviews, note that its marginal variance is

$$\text{Var}(Y_r) = \tau_\theta^2 + \tau_b^2 + \sigma^2.$$

For posterior draw s , define

$$V^{(s)} = \left(\tau_\theta^{(s)}\right)^2 + \left(\tau_b^{(s)}\right)^2 + \left(\sigma^{(s)}\right)^2.$$

The corresponding variance shares are

$$\rho_{\text{proposal}}^{(s)} = \frac{\left(\tau_\theta^{(s)}\right)^2}{V^{(s)}}, \quad \rho_{\text{reviewer}}^{(s)} = \frac{\left(\tau_b^{(s)}\right)^2}{V^{(s)}}, \quad \rho_{\text{residual}}^{(s)} = \frac{\left(\sigma^{(s)}\right)^2}{V^{(s)}}.$$

For the overall-grade V1 model, the posterior medians of these shares were 0.050, 0.026 and 0.896, respectively. The corresponding 95% credible intervals were approximately [0.0002, 0.289], [0.0001, 0.240] and [0.615, 0.995].

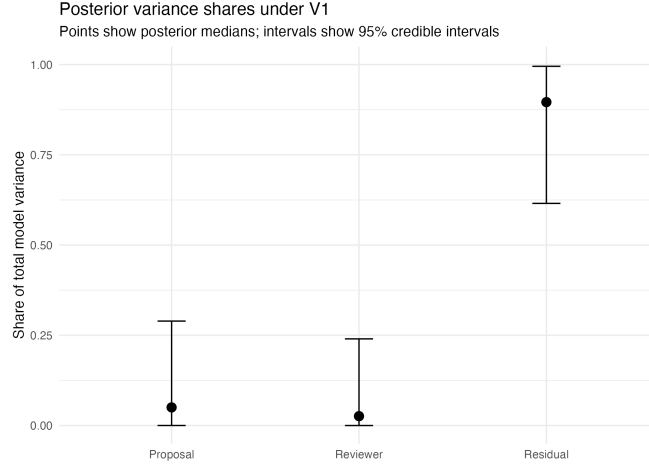


Figure 5: Posterior variance shares under V1. Points show posterior medians and vertical intervals show 95% credible intervals.

The proposal share also has a reliability interpretation. For two ratings of the same proposal i made by different reviewers j and k ,

$$\text{Cov}(Y_{ij}, Y_{ik}) = \text{Var}(\theta_i) = \tau_\theta^2,$$

because the two observations share the same proposal effect but have different reviewer and residual terms. Hence,

$$\text{Cor}(Y_{ij}, Y_{ik}) = \frac{\tau_\theta^2}{\tau_\theta^2 + \tau_b^2 + \sigma^2} = \rho_{\text{proposal}}.$$

The proposal variance share is therefore also the model-implied intraproposal correlation under V1, following the cross-classified reliability interpretation of Jayasinghe et al. (2003). Its posterior median of approximately 0.05 indicates weak model-implied correlation between two ratings of the same proposal. The dominant residual share is consistent with the broad posterior rank distributions in Chapter 4, although this decomposition alone does not identify the sparse assignment design as their cause.

For comparison, the same variance decomposition was calculated for the V1 fit using the equal-weight criterion average A_r as the response.

Response	Proposal share	Reviewer share	Residual share
Holistic overall grade	0.050	0.026	0.896
Equal-weight criterion average	0.192	0.046	0.733

For every posterior draw, the three variance shares sum to one. The values reported in the table are the medians of the three posterior distributions calculated separately, so their reported medians need not sum exactly to one.

The criterion-average model attributes a larger share of its fitted variation to differences between proposals and a smaller share to residual review-level variation than the holistic-grade model. However, the two decompositions concern different response variables: A_r averages four criterion scores and has lower observed dispersion than the holistic grade. The larger proposal share should therefore not be interpreted as evidence that the equal-weight criterion average is intrinsically a better evaluation measure.

5.3 Posterior predictive assessment

Posterior predictive checks assessed whether V1 reproduces the observed location, dispersion and reviewer disagreement. For posterior draw s and review r ,

$$Y_r^{\text{rep},(s)} \sim \mathcal{N}\left(\bar{Y} + \theta_{p(r)}^{(s)} + b_{a(r)}^{(s)}, (\sigma^{(s)})^2\right).$$

Across 5,000 replicated datasets, the observed mean grade of 3.476 was close to the posterior predictive mean of 3.474, and the observed standard deviation of 0.950 was close to the replicated mean of 0.975. The observed mean absolute disagreement between the two reviews of a proposal was 1.095, compared with a posterior predictive mean of 1.064; approximately 40% of replicated datasets produced at least as much mean disagreement as observed.

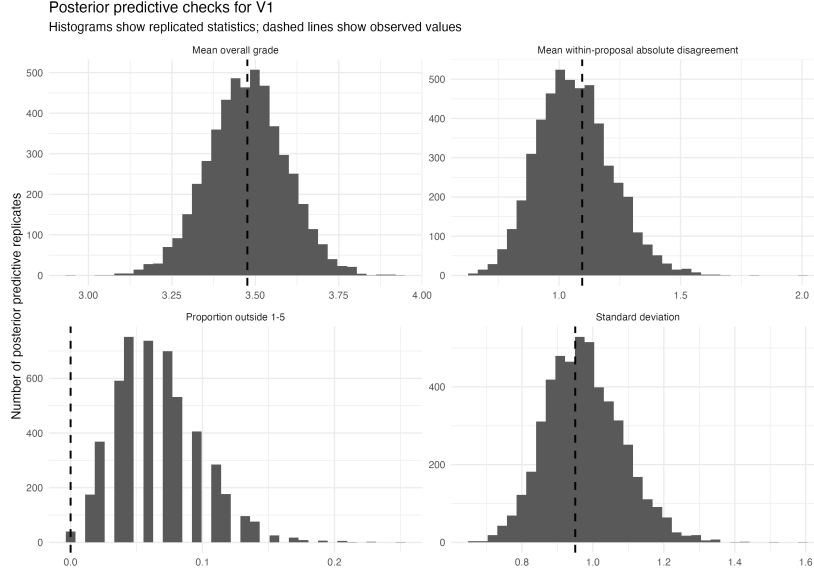


Figure 6: Posterior predictive checks for V1. Histograms show the distributions of statistics calculated from replicated datasets, while dashed vertical lines show the corresponding observed values.

A more detailed disagreement check showed that the low posterior predictive probability for disagreements of at least two score points is largely explained by comparing continuous replicated scores with integer-valued observed grades. After mapping the replicated grades to the integer scale from 1 to 5, the apparent mismatch became much smaller: the posterior predictive probability increased from 0.0088 to 0.272. The derivation and additional disagreement checks are given in Section 9.4.

The main model-form limitation is visible in the lower-left panel of Figure 6. Because the Gaussian likelihood is unbounded, an average of 6.6% of replicated grades fall below 1 or above 5. V1 is therefore retained as a continuous approximation for latent proposal effects and ranking uncertainty, not as a literal generative model for valid future score categories.

The corresponding posterior predictive intervals are reported in Section 9.5.

5.4 Prior sensitivity

The baseline scale priors

$$\sigma, \tau_\theta, \tau_b \sim \text{Uniform}(0, 2)$$

were replaced by independent $\text{HalfNormal}(0, 1)$ priors while keeping the likelihood, data and MCMC settings unchanged. The ranking conclusions were essentially unchanged: the Pearson correlation between expected ranks from the two prior specifications was 0.99994, the mean absolute expected-rank change was 0.093 positions and the maximum was 0.283. The median width of the 90% posterior rank intervals was 37 under both specifications. Posterior median variance shares were also nearly identical: the proposal share changed from 0.050 to 0.046, the reviewer share from 0.026 to 0.025 and the residual share from 0.896 to 0.902. The corresponding expected-rank figure and additional numerical details are provided in Section 9.6.

5.5 Validation conclusion

The validation provides no strong indication that the main V1 ranking conclusions are driven by computational instability, the tested scale priors or a failure to reproduce the observed mean, dispersion and reviewer-disagreement structure. The fitted model attributes most variation to residual review-level variation, and the exact proposal ranks remain weakly identified.

The principal model-form caveat is the continuous Gaussian treatment of the five-point score, which permits replicated values outside the admissible range. Subject to this approximation and the non-random sparse assignment design, V1 is retained as the working model for the review-design analysis in Chapter 6.

6 Graph-based review design

6.1 Motivation and scope

6.2 Reviewer–proposal assignment structure

6.3 Statistical relevance of the assignment structure

6.4 Quantifying uncertainty for review design

6.5 Additional review scenarios

6.6 Comparison of review-design approaches

6.7 Generalization beyond the CRS design

7 Discussion and recommendations

7.1 Answers to the research questions

7.2 Interpretation of the main findings

7.3 Implications for CRS

7.4 Ranking uncertainty and decision making

7.5 Limitations

7.6 Future work

8 Conclusion

9 Supplementary analyses

9.1 Correlation definitions

For criterion c , the sample Pearson correlation with the holistic overall grade is

$$\text{Cor}_P(C_{\cdot c}, Y) = \frac{\sum_{r=1}^n (C_{rc} - \bar{C}_c)(Y_r - \bar{Y})}{\sqrt{\sum_{r=1}^n (C_{rc} - \bar{C}_c)^2 \sum_{r=1}^n (Y_r - \bar{Y})^2}},$$

where $n = 84$, $C_{\cdot c} = (C_{1c}, \dots, C_{nc})$ and $Y = (Y_1, \dots, Y_n)$. The corresponding sample Spearman correlation is

$$\text{Cor}_S(C_{\cdot c}, Y) = \text{Cor}_P(R(C_{\cdot c}), R(Y)),$$

where $R(\cdot)$ denotes the rank transformation and average ranks are used for ties.

9.2 V0 reference-model details

Under V0, P022 had posterior-mean rank 1 and expected rank 11.85. P023, P032 and P041 occupied posterior-mean ranks 2, 3 and 4, with expected ranks 15.45, 15.49 and 16.50. Their 90% posterior rank intervals were 1–35, 1–37, 1–37 and 2–38, respectively. The maximum PSRF across monitored V0 parameters was 1.0077 and the minimum effective sample size was 501.

For completeness, Figure 7 shows the comparison between the observed-score rank, posterior-mean rank and expected rank under V0.

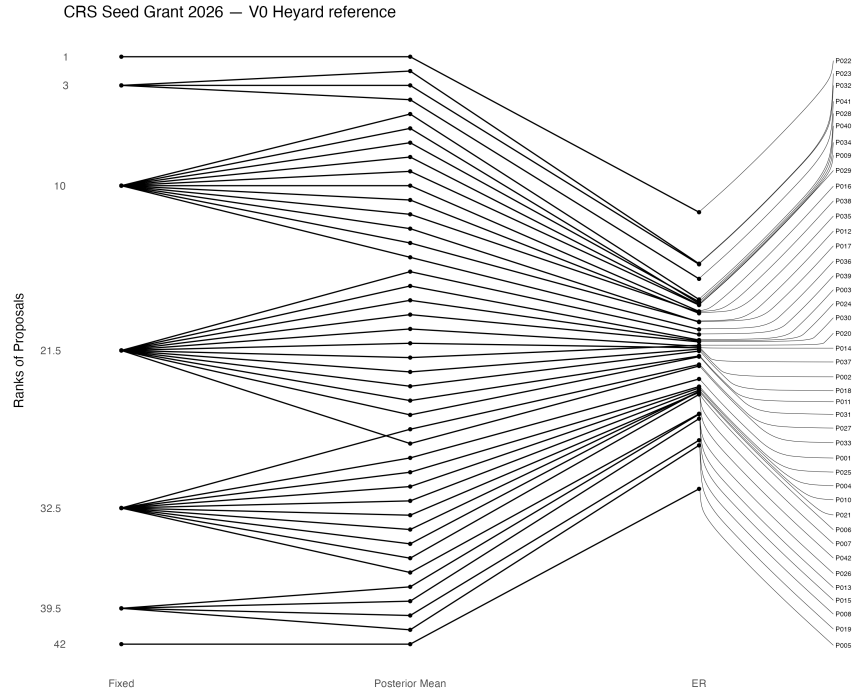


Figure 7: Proposal rankings under the observed-score ranking, posterior-mean ranking and posterior expected rank for the V0 Heyard reference model.

9.3 V1 MCMC trace plots

Figure 8 shows the final 5,000 stored iterations from each V1 chain, plotting every fifth value for readability. Numerical diagnostics in the main text are based on the full posterior sample.

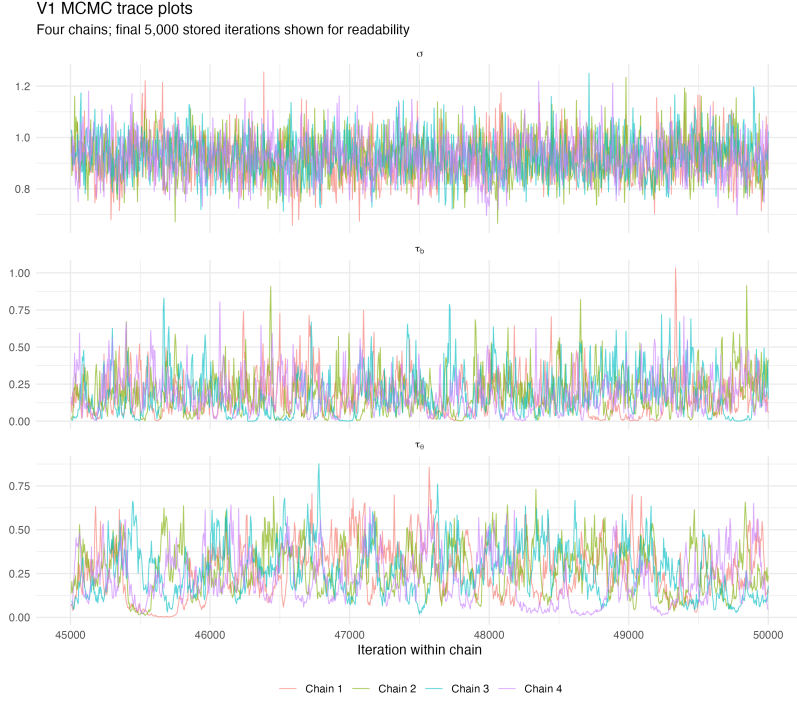


Figure 8: Trace plots for the V1 scale parameters τ_θ , τ_b , and σ .

9.4 Detailed reviewer-disagreement diagnostic

Consider proposal i , reviewed by reviewers j and k . Under V1,

$$Y_{ij} - Y_{ik} = b_j - b_k + \varepsilon_{ij} - \varepsilon_{ik},$$

so the proposal effect cancels and

$$Y_{ij} - Y_{ik} \mid b_j, b_k, \sigma \sim \mathcal{N}(b_j - b_k, 2\sigma^2).$$

Taking the absolute value gives a folded normal distribution ([Tsagris et al. 2014](#)). If $X \sim \mathcal{N}(\mu, s^2)$, then

$$\mathbb{E}|X| = s\sqrt{\frac{2}{\pi}} \exp\left(-\frac{\mu^2}{2s^2}\right) + |\mu| \left[2\Phi\left(\frac{|\mu|}{s}\right) - 1\right],$$

where Φ is the standard normal cumulative distribution function. In the present setting, $\mu = b_j - b_k$ and $s = \sqrt{2}\sigma$. Evaluating this expression for the observed reviewer pairs and posterior draws gave analytic posterior mean disagreement

1.066, almost identical to the simulated posterior predictive mean 1.064. The posterior predictive probability of mean disagreement at least as large as the observed value 1.095 was 0.401.

Most of the model-implied disagreement is associated with residual review-level variation. If the two reviewer effects are equal, the expected absolute disagreement simplifies to

$$E|Y_{ij} - Y_{ik}| = \frac{2\sigma}{\sqrt{\pi}},$$

which averaged 1.043 over the posterior. Allowing the fitted reviewer effects to differ increased the expected mean disagreement by only about 0.023 on average.

A threshold-based check initially suggested underprediction of moderate disagreement. Observed absolute disagreement was 0 for 10 proposals, 1 for 19 proposals, 2 for 12 proposals and 3 for one proposal. Thus 13 of 42 proposals had disagreement of at least two score points, giving

$$T_{\text{obs}} = \frac{13}{42} \approx 0.310.$$

With continuous Gaussian replicated grades, the posterior predictive mean of the corresponding proportion was 0.134 and

$$\mathbb{P}(T_{\text{cont}}^{\text{rep}} \geq T_{\text{obs}} \mid Y) = 0.0088.$$

Because observed scores are integer-valued, a diagnostic version of each replicated score was mapped to the 1–5 grid,

$$\tilde{Y}_r^{\text{rep}} = \min \{5, \max \{1, \text{round}(Y_r^{\text{rep}})\}\}.$$

After this transformation, the posterior predictive mean proportion with disagreement of at least two increased to 0.255 and

$$\mathbb{P}(T_{\text{disc}}^{\text{rep}} \geq T_{\text{obs}} \mid Y) = 0.272.$$

The broader disagreement pattern is summarized in Figure 9. The observed mean, median, 90th percentile, maximum and proportion of disagreements of at least three score points are all compatible with their posterior predictive distributions. The clearest departure is the frequency of disagreements of at least two score points, which is higher in the observed data than in most continuous replicated datasets. As shown above, this discrepancy is substantially reduced once replicated grades are placed on the same integer-valued 1–5 scale as the observations.

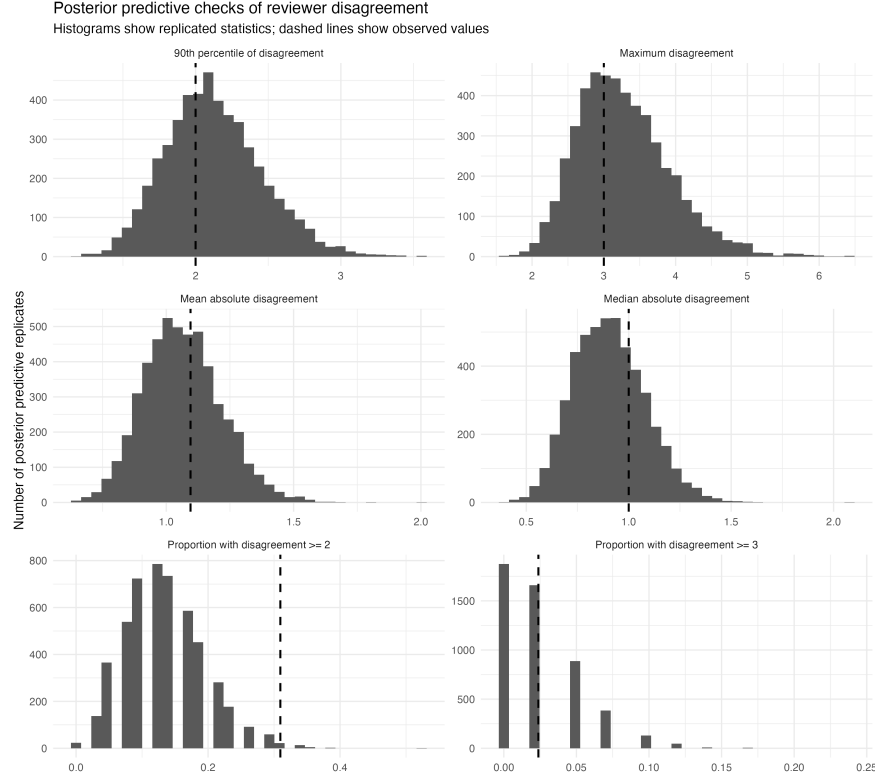


Figure 9: Posterior predictive checks for several summaries of reviewer disagreement under V1. Histograms show statistics from replicated datasets and dashed vertical lines show the corresponding observed values.

9.5 Additional posterior predictive summaries

Statistic	Observed	Replicated mean	2.5%	97.5%
Mean overall grade	3.476	3.474	3.241	3.704
Standard deviation	0.950	0.975	0.781	1.201
Mean absolute disagreement	1.095	1.064	0.790	1.386
Proportion outside 1–5	0.000	0.066	0.012	0.143

The intervals are the 2.5th and 97.5th percentiles of the corresponding statistics across the 5,000 posterior predictive datasets.

9.6 Prior-sensitivity details

Under the HalfNormal(0,1) scale priors, the posterior medians of τ_θ , τ_b and σ were 0.209, 0.154 and 0.916, compared with 0.220, 0.157 and 0.920 under

the baseline specification. The maximum PSRF was 1.0011 and the minimum effective sample size was 1,306.

The Spearman correlation between posterior-mean ranks under the two prior specifications was 0.9990. Six proposals changed integer posterior-mean rank, but the changes were small: the mean absolute change was 0.19 positions and the maximum was two positions. The mean widths of the 90% posterior rank intervals were 36.74 and 36.83 under the baseline and half-normal specifications, respectively.

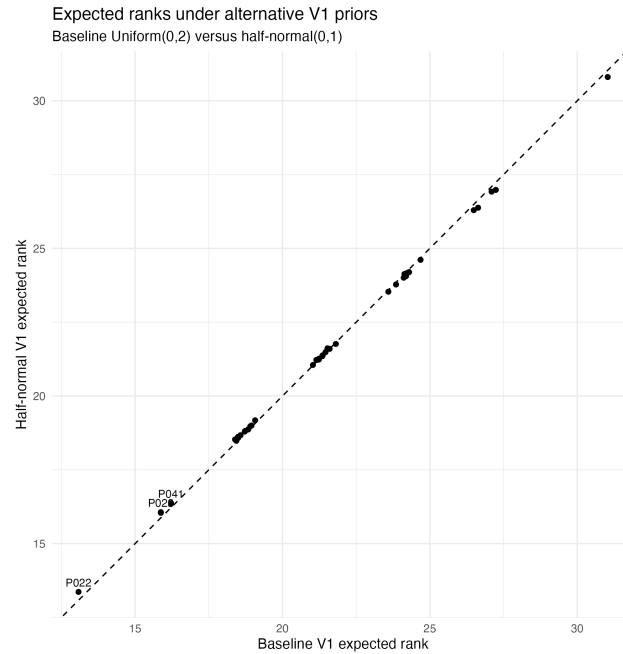


Figure 10: Posterior expected ranks under the baseline V1 prior specification and the half-normal prior-sensitivity specification. The dashed identity line represents identical expected ranks under both models.

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